

**DESIGN OF VERTICAL AXIS WIND TURBINE USING AXIAL FLUX
MOTORS**

PROJECT REPORT

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*Submitted in partial fulfilment of the
requirements of the award of Bachelor of Technology Degree
in Electrical and Electronics Engineering of the
APJ Abdul Kalam Technological University (KTU)*

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CERTIFICATE

*This is to certify that this report entitled “**DESIGN OF VERTICAL AXIS WIND TURBINE USING AXIAL FLUX MOTORS**” is a bonafide record of the project presented by **JOANA J JOHN** (VII Semester, MBT19EE046), **RIYA RAJESH** (VII Semester, MBT19EE062), **SANJAY JS** (VII Semester, MBT19EE0), **SUJIN REJI** (VII Semester, MBT19EE079) towards the partial fulfilment of the requirements for the award of B.Tech Degree in Electrical and Electronics Engineering of the APJ Abdul Kalam Technological University (KTU) during the year 2022.*

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ABSTRACT

Today with the depletion of primary energy in the form of petroleum and coal, which has led to soaring prices of oil and coal, resulting in swelling in electricity costs in the world. The swelling is also because 85% of electricity fuel is petroleum and coal and added by the development of technology in this era, which will require more electricity resources to supply the needs and demands of consumers to the electricity. Therefore, switching to renewable resources will be needed. It has been found that after the comparison of axial fluxes and permanent radial magnet generators, axial-flux permanent magnets have shown the best results, which are axial-flux generators cheaper than radial-fluxes. So, the importance of the availability of alternative power plants using potential energy from nature will be urgently needed for now and beyond. An axial flux permanent magnet machine is designed to operate as a generator in a small-scale wind-power application. The machine is realized by using consecrated stator winding with open slots and surface mounted permanent magnets on rotor disk. Such a novel generator structure is simple to construct and its performance is good; it offers sinusoidal back-emf waveform, low torque ripple and high efficiency.

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NOMENCLATURE

- HAWT - Horizontal Axis Wind Turbines
- VAWT - Vertical Axis Wind Turbines
- CFD - Computational Fluid Dynamics
- FEA -
- EMF – Electro Motive Force

1. INTRODUCTION

1.1. Introduction

In recent era, research and development activities in the field of renewable energy, especially wind and solar, have been considerably increased, due to the worldwide energy crisis and high global emission. A wind turbine converts the kinetic energy in wind into mechanical energy, which will be reflected on its axis. To convert this mechanical energy into electrical energy, the turbine has to be coupled to an electrical generator, becoming a wind turbine. To perform this task, several types of turbines are used, differentiated into two types by the position of its axis of rotation with respect to the surface on which it is fixed, and can be then Horizontal Axis Wind Turbines (HAWT), the most common turbines, used mostly in wind farms for high-energy production, and Vertical Axis Wind Turbines (VAWT), latter less common and currently are the subject of numerous studies and developments of new models. The horizontal axis wind turbine cannot be used for household purpose. So, Savonius vertical axis wind turbine can be better option as it operates in low wind condition also. The choice for this model is to showcase its efficiency in varying wind conditions as compared to the traditional horizontal axis wind turbine and contribute to its steady growing popularity for the purpose of mass utilization in the near future as a reliable source of power generation.

1.2. EXISTING TECHNOLOGY

1.2.1. CLASSIFICATION OF WIND TURBINES

Most wind turbines fall into one of two general categories: horizontal axis (HAWT) and vertical axis (VAWT). Each can be further divided into small and large wind turbines. Wind turbines can be classified as utility scale and small scale and then into a horizontal axis (HAWTs) and vertical axis versions (VAWTs).

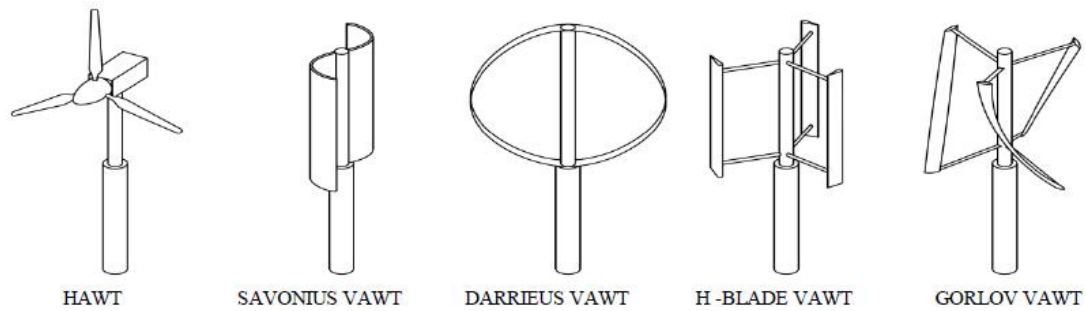


Figure 1-1: Different types of wind turbines

The technology for traditional horizontal axis wind turbines (HAWTs) have been in development for more than a hundred years. This technology includes blades and their manufacturing, gear boxes and their manufacturing technology, pitch angle devices and their technology, and so on. The technologies are very mature. Betz's Law, which defines the maximum amount of energy that a HAWT can extract from the wind, is based on a single disk (the rotor) moving in a two-dimensional space.

1.2.2. VERTICAL AXIS WIND TURBINES



Figure 1-2: Vertical Axis Wind Turbines

Within the VAWT's, there are two types, differentiated by the morphology of their blades.

Savonius type or S turbine, created in 1922, is composed of two circular cross-sections blades, one placed concavely and one convexly towards the wind position and vertically arranged along an axis of rotation with the particular features

that the blades are overlapping in the vicinity of the rotation axis, making the effect of support by exchanging the flow between both blades, thus achieving assistance for the starting factor that receive both blades to be positive in the wind direction and thereby getting movement.

The second type of VAWT is the Darrieus type, which was created in 1931, has the particularity that the blades are aerodynamically shaped, usually NACA style, with different layouts and with a certain distance from the axis of rotation.

1.3. OBJECTIVE OF THE PROJECT

Due to the increase of the environmental concern, the electrical power generation from renewable energy sources, such as wind, is increasing and adopted. Wind is suitable for either grid integration or as a stand-alone system used for remote areas for illuminating purposes or in charging batteries. The VAWT is controlled to operate at maximum mechanical power output, thus extracting maximum power from the wind into electricity. The output voltage of DC-DC converter is controlled to be constant at any wind speed for battery charging applications. For small wind speeds, the output voltage is boosted and for large wind speeds, the output voltage is bucked. The use of permanent magnets in a coreless axial flux generator makes the construction process easier and suitable for locally manufactured open hardware wind power application. The machine is realized by using concentrated stator winding with open slots and surface mounted permanent magnets on rotor disk, such that it offers sinusoidal back emf waveform, low torque ripple and high efficiency because of the larger diameter of spinning rotor which turns alongside the stator instead of inside it like the conventional radial flux generators.

1.4. METHODOLOGY OF THE PROJECT

- For the generator design: The Body of the designed generator is modelled using 3D modelling and printed using 3D printing.
- The coil is wound upon the printed stator part and the permanent magnets are fixed upon the respective rotor part.

- The developed generator is then attached to the shaft and the Savonius type blades are mounted to the shaft.
- The generated AC supply is rectified into DC and stored into battery for further applications.
- The MATLAB simulation for the same will be done under various wind conditions.

1.5. Organization of the report

The report is divided into chapters. The first chapter is introductory in nature giving the The second chapter describes the and so on.

1.6. Summary

This chapter gives an introduction of the project report and indicates how the whole report is arranged as chapters. A detailed review of the available literature is given in the next chapter.

2. LITERATURE REVIEW

2.1. PAPER 1

A. Parviainen. J, Pyrhönen. P, Kontkanen, “Axial Flux Permanent Magnet Generator with Concentrated Winding for Small Wind Power Applications”, Electric Machines and Drives, *IEEE International Conference, 2005.*

This study describes an axial flux permanent magnet machine that is intended to function as a generator in small-scale wind power applications. Surface-mounted permanent magnets are used on the rotor disc and a stator winding with open spaces to create the machine. A straightforward model generator structure with good and effective performance.

A 1.6 kW axial flux permanent magnet generator for direct-drive wind-power applications was described. The proposed concentrated winding concept with open slots makes the generator manufacturing easy and it offers good performance. The obtained phase voltage and current is practically pure sine wave, and the torque ripple is low. The efficiency of the machine is also good. As an author's opinion, the mechanical structure of the machine proposed is simpler and cheaper to construct than the other axial flux machine configurations with laminated stator cores

2.2. PAPER 2

Ahmad M.Eid, Mazen Abdel-Salamand M.Tharwat Abdel-Rahman, “Vertical-axis wind turbine modeling and performance with axial-flux permanent magnet generator for battery charging applications”, *The Eleventh International Middle East Power System Conference (MEPCON' 2006), December 2006.*

Using the PSIM6 software programme, modelling, and simulation of a 1 kW VAWT are carried out in this research. The VAWT's transient and steady

state simulations at various wind speeds are explained. The VAWT is programmed to run at maximum mechanical output, converting the most wind energy possible into electricity. For battery charging applications, the DC-DC converter's output voltage is adjusted to remain constant regardless of wind speed. The output voltage is increased for low wind speeds and decreased for high wind speeds.

The project work assisted in developing a thorough understanding of wind energy harvesting methods and the enormous potential in the industry. The main idea at first was to create a wind turbine that is effective in terms of minimal energy loss. A sustainable and environmentally friendly energy production system was urgently needed in light of the rising pollution rates, frightening environmental dangers, such as global warming, and other worrying phenomena. Additionally, the model was designed to run domestically and even in windy conditions. The area's disregard for creating localised energy-producing units was another startling aspect; transmission losses are one of the main sources of energy loss.

2.3. PAPER 3

Kostas Latoufis, Georgios Messinis, Panos Kotsampopoulos, “Axial Flux Permanent Magnet Generator Design for Low-Cost Manufacturing of Small Wind Turbines”, 7th International Conference on PV Hybrids and Mini Grids. National Technical University of Athens. July 6, 2016.

For low-cost electrification applications, the design and production procedures for axial flux permanent magnet generators are presented. The method is based on already published open-source design and building instructions, and it describes a systematic method for designing generators with different nominal powers that may be connected to the grid or used directly to charge batteries. Simple methods and techniques are utilised to save costs.

2.4. PAPER 4

Ashwin Dhote, Prof. Vaibhav Bankar "Design, Analysis and Fabrication of Savonius Vertical Axis Wind Turbine", *International Research Journal of Engineering and Technology (IRJET) Volume: 02 Issue: 03 June-2015.*

To produce electrical energy from wind energy, the vertical axis Savonius wind turbine's design and analysis are presented in this work. The Savonius rotor's diameter and height are 2 and 4 metres, respectively. Using the software Solid Works, the Savonius rotor blade 3D model was created. This paper presents structural Finite Element Analysis and Computational Fluid Dynamics (CFD) analysis. The pressure difference between the rotor blades concaves and convex surfaces was determined using CFD analysis, and the structural response of the blade was determined using FEA.

2.5. PAPER 5

M.Zemamoua, M.Aggoura , A.Toumi, "Review of Savonius Wind Turbine Design and Performance", *4th International Conference on Power and Energy Systems Engineering, CPSE 2017, 25-29 September 2017, Berlin, Germany*

The world has adopted an energy transition policy, which entails replacing fossil fuels with renewable energies to reduce CO₂ emissions; however, the main challenge is to develop a wind turbine with a simple design, a relatively low operating speed, and independent wind directions. The Savonius rotor initially seems to be promising in these circumstances, but it has significant drawbacks: low efficiency and high negative torque. Numerous scientific researchers are working to optimise the effects of various geometric parameters and create new designs to enhance the Savonius turbine's capabilities. For traditional Savonius rotors, the power coefficient values vary from 0.1 to 0.25. The addition of numerous extra sets results in new designs with improved performance

2.6. PAPER 6

Ilham Satrio Utomoa, Dominicus Danardono Dwi Prija Tjahjanab, Syamsul Hadic "Experimental Studies of Savonius Wind Turbines with Variations Sizes and Fin Numbers Towards Performance" AIP Conference Proceedings 1931, 030041 (2018); <https://doi.org/10.1063/1.5024100>, 09 February 2018.

In Indonesia, little renewable energy is still used. particularly the usage of wind power. One wind turbine that can operate in light winds is the Savonius. Savonius wind turbines are still inefficient, nevertheless. Therefore, modifying is required. By generating a flow that can penetrate the overlap ratio of the gap, modifications made with the fin are anticipated to improve the positive drag force. This study was carried out utilising an experimental methodology. Power generator, power coefficient, and torque coefficient are some of the parameters produced by the experiment. The experimental data will be gathered by varying the horizontal fin area while the wind speed ranges from 3 m/s to 4,85 m/s. According to experimental findings, the Savonius wind turbine may perform 11% better when a fin is added. A fin with a diameter of 115 mm can deliver the Savonius wind turbine's full performance.

3. WORKING

3.1. INTRODUCTION

This chapter discusses about the working of the vertical axis wind turbine using the axial flux motor.

3.2. WORKING

Due to the curvature shape, the rotor will have minimal dragging when they move in the opposite direction of the wind, whereas they experience maximum drag while moving in the same direction of the wind. This variation in the dragging creates the spinning movement of the turbine. When the turbine spins, the rotor (Permanent magnet side) of the generator which is connected to this is also rotated. Thus, an EMF is induced in the stator coil. This generated AC supply is given to the rectifier and converted into DC supply, and it is stored into battery for further applications.

3.3. SUMMARY

This chapter discusses about the working of the proposed project detailed.

4. DESIGN DETAILS

4.1. INTRODUCTION

This chapter describes about the design details of the wind turbine with corresponding detailed figures.

4.2. DESIGN DETAILS

4.2.1. GENERATOR

- The generator body is designed and printed using 3D printing.
- Coil winding is wound upon coil slot on the stator side.
- Permanent magnets are fixed on the respective spaces on rotor parts.
- Both the stator and two rotors are connected to the shaft by bearings.
- An example figure is shown here:

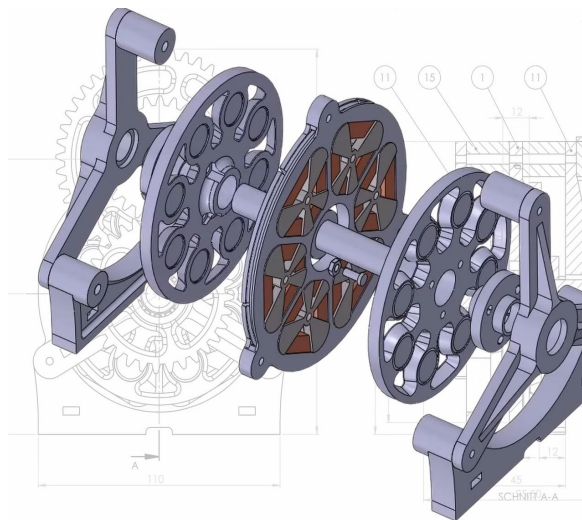


Figure 4- 1:Generator design

4.2.2. TURBINE

- Savonius type rotors are used.
- For that a PVC semi-circular blades are made out.

- Similarly, 6 blades are made and fixed to shaft like in the figure given below.
- They are fixed by nut and bolt to the shaft.



Figure 4- 2 : Turbine Design

5. ADVANTAGES AND DISADVANTAGES

5.1. INTRODUCTION

Vertical axis type turbine is unusual and its application for obtaining useful energy from air stream is an alternative to the use of conventional wind turbines in now a days. Over the years, numerous adaptations for this device were proposed. The variety of possible configurations of the rotor is an advantage in using such machine. However, it also has certain disadvantages it can be used for small scale purposes.

5.2. ADVANTAGES

- Independence on wind direction, no additional control mechanisms are required
- Ability to operate in a wide range of wind conditions (turbulence level, wind speed)
- Electrical equipment can be placed at ground level
- Low noise emission
- High starting torque
- Compact size
- Simple and cheap construction.

5.3. DISADVANTAGES

- Less Rotation Efficiency
- Lower Available Wind Speed.
- Component Wear-down.
- Less Efficiency.
- Self-Starting Mechanism.

Despite these disadvantages, we believe vertical axis wind turbines are still worth investing in and developing. It is certain that this type of

wind turbine has the potential to not only establish its niche market, but also provides alternatives to horizontal axis models where they do not fit.

5.4. SUMMARY

This chapter gives the idea about the advantages and disadvantages of vertical axis wind turbine.

6. COMPONENTS

6.1. INTRODUCTION

This chapter gives the details about the components required for this project.

6.2. REQUIRED ITEMS

6.2.1. MAGNETS (GENERATOR ROTOR)

The magnets used are N52 Neodymium Magnets (Strong). It is the heart of the generator. It's the permanent magnet on the rotor plates.



Figure 6- 1: Neodymium Magnets

6.2.2. COIL WINDING

Copper coil of 0.50mm thickness is used for the stator winding. Copper coil is wound upon the stator.



Figure 6- 2: Copper Coil

6.2.3. SHAFT

Cylindrical metal rods are used for shaft purpose.



Figure 6- 3: Steel shaft

6.2.4. TURBINE

Turbine is an important part of this project. Here we use a savinous type turbine, which is made up of PVC.



Figure 6- 4: PVC turbines

6.2.5. AC-DC CONVERTER

An AC-DC converter is used to convert the generated AC into DC.



Figure 6- 5: AC-DC Converter

6.2.6. DC-DC CONVERTER

A DC-DC converter is used to efficiently regulate the produced DC.



Figure 6- 6: DC-DC Converter

6.2.7. BATTERY

Battery is used to store and supply the generated DC.



Figure 6- 7: Battery

7. ESTIMATE COST

Items Used	Amount
Magnets (Generator rotor)	11200/-(16*700 per piece)
Coil Winding	5000/-
Shaft	1000/-
PVC Turbine	2000/-
AC-DC Converter	500/-
DC-DC Converter	500/-
Battery	5000/-
Total	35200/-

Table 7- 1:Cost Estimation

8. WORK PLAN

SL NO	MONTH	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY
1	CREATING SOLUTION FOR OBSERVED PROBLEM	✓	✓						
2	LITERATURE REVIEW			✓					
3	PHASE ZERO PRESENTATION		✓						
4	PHASE ONE PRESENTATION			✓					
5	SELECTION OF COMPONENTS			✓					
6	IMPLEMENTATION OF THE PROJECT			✓	✓	✓	✓		
7	TESTING							✓	✓
8	FINAL PROJECT								✓
9	DOCUMENTATION								✓

Table 8- 1: Work Plan

9. CONCLUSION

There are number of sources for generation of power but in the recent years wind energy shown its potential as the clean source of energy and contributing to the high energy demands of the world. Vertical axis wind turbine is the best option for the area which are under load sheading. The output from the turbine is use to charge a heavy-duty battery. This can be beneficial than inverter back up.

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